Ulaanbaatar, Mongolia

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EDUCATION

Aug 2023 — Present

Bachelor of Computer Science, Mongolian University of Science and Technology

GPA: 3.52

SKILLS

Languages: C++, Python, Java, JavaScript, SQL

Frameworks & Libraries: React, Vue.js, Node.js, Express, Spring Boot, PyTorch

Tools & Platforms: Git, Docker, PostgreSQL, MongoDB, ESP32, Linux, 3D Printing (FDM)

SOFTWARE PROJECTS

POS & Branch Server System

Utilized: Spring Boot, Java, PostgreSQL, JPA, Docker, K6

- Designed and built a REST API backend for a retail POS system supporting branch-level sales recording, product lookups, and local-first operation during network outages with subsequent sync to the branch server
- Implemented offline-first architecture allowing POS terminals to cache transactions locally and synchronize data once connectivity is restored, ensuring zero data loss
- Load-tested the API using K6 against 34,500 requests at 285 RPS — achieved 23ms average response time, 51ms P95 latency, and 0% error rate across all test scenarios
- Containerized the application with Docker for consistent local and deployment environments

3D Model Ecommerce Platform

Utilized: Vue.js, Node.js, Express, MongoDB, JWT

- Built a full-stack marketplace for selling and distributing STL files, covering the complete user flow from browsing to purchase and file download
- Engineered RESTful backend APIs for user account management, product listings, and secure file delivery with JWT-based authentication
- Implemented dynamic content rendering and client-side routing for a smooth single-page application experience

Synthetic Human Face Generator

Utilized: Python, PyTorch, GAN, NumPy, Matplotlib

- Trained a Generative Adversarial Network (GAN) on an ethically sourced facial dataset to synthesize realistic human faces
- Engineered the end-to-end training pipeline including data preprocessing, augmentation, and model checkpointing for stability
- Experimented with training techniques such as learning rate scheduling and gradient clipping to reduce mode collapse and improve output diversity
- Evaluated generated outputs qualitatively and quantitatively, iterating on architecture and hyperparameters to improve realism

Hand Gesture-Controlled Embedded Device

Utilized: Python, OpenCV, MediaPipe, C++, ESP32, Wi-Fi

- Developed a real-time hand gesture recognition system using MediaPipe and OpenCV, achieving reliable detection across variable lighting conditions
- Built a wireless communication layer transmitting gesture-derived control signals from a PC to an ESP32 microcontroller over Wi-Fi
- Programmed the ESP32 in C++ to interpret incoming signals and drive connected peripherals, bridging computer vision with hardware control
- Designed robust gesture classification logic to minimize false triggers and ensure consistent device response

HOBBIES

Cars, Cycling